



SCHOOL OF COMPUTING
FACULTY OF ENGINEERING
UNIVERSITI TEKNOLOGI MALAYSIA

FINAL YEAR PROJECT (PSM) STUDENT LOG BOOK

PROJECT TITLE

DESIGN AND DEPLOYMENT OF A SECURE CLOUD-BASED
PATIENT DATA MANAGEMENT SYSTEM USING A THREE-TIER
ARCHITECTURE ON AWS

STUDENT INFO

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Department : IS / SE / **CS**

SUPERVISOR INFO

Main Supervisor : Johan Mohamed Sharif

Email : johan@utm

Co-Supervisor (if applicable):.....

STUDENT LOG BOOK NOTES

- 1) This log book need to be used by a PSM 1 and PSM 2 student for a purpose of reporting all progress of their PSM projects.
- 2) It is the responsibility of the student that this log book is **kept up to date** and that the student complies with the Supervisor's suggestions and recommendations as noted by the student in the log book and approved by the Supervisor.
- 3) Student **MUST** makes a regular meeting with a supervisor at least **ONCE** in **TWO WEEKS**. A minimum number of meetings for PSM student-supervisor meeting are 6 times per semester.
- 4) A PSM Department Coordinator and supervisor have a right for not to allow the student to present their project if this regulation is not complied.
- 5) This log book has to be submitted together with the proposal report (for PSM 1) and the final thesis report (for PSM 2) as stated in the PSM Activities Calendar.
- 6) The Faculty of Computing reserves right not to accept thesis for examination if this log book is not properly documented.



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PSM LOG BOOK

SEMESTER: 20252026-2 TYPE: **PSM 1** / PSM 2

Date :26/03/2026 Meeting: **1** / 2 / 3 / 4 / 5/ 6 / 7

Student
(Meeting Minute/
Achievements/
Activities)

: - **Title:** Discussed PSM1 Proposal title and system development
 - **Stakeholder Analysis:** Defined the primary stakeholder, Alamin Clinic (Saudi Arabia), and used their recent ransomware attack as a core case study for system design
 - **Proposed Architecture:** Finalized and Searched the design of a Three-Tier Clinic Record Management System on AWS VPC

Supervisor
(Suggestion&
Comments)

: - **Scope Management:** Advised focusing on core functional modules
 - **Validation Metrics:** Encouraged putting a metrics to validate the system
 - **Objectives:** Advised on the objective academic writing

Next Meeting Plan

: Review PSM1 proposal Changes

Supervisor's Signature

..... Date:



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PSM LOG BOOK

SEMESTER: 20252026-2 TYPE: **PSM 1** / PSM 2

Date : 02/04/2026 Meeting: 1 / **2** / 3 / 4 / 5 / 6 / 7

Student : - **Edited** Objectives based on comments
(Meeting Minute/ - **Specified** and narrowed the scope
Achievements/ - **Added** more on Problem statements and proposed solution
Activities)

Supervisor : - **Recommended** to add Literal review on existing works
(Suggestion& - **Advised** on referencing with no more than 5 to 10 years
Comments) - **Advised** on comparing existing and proposed system

Next Meeting : Review Edits based on comments and Processed with Progress 1 meeting
Plan

Supervisor's
Signature Date:



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PSM LOG BOOK

SEMESTER: 20252026-2 **TYPE:** **PSM 1** / PSM 2

Date : 10/04/2026 **Meeting:** 1 / 2 / **3** / 4 / 5 / 6 / 7

Student : - **Meted** to briefly review the final proposal outcome
(Meeting Minute/ - **Presented** and no farther comments given
Achievements/
Activities)
Supervisor : N\A
(Suggestion&
Comments)

Next Meeting : Progress 1
Plan

Supervisor's
Signature **Date:**



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PSM LOG BOOK

SEMESTER: 20252026 **TYPE:** **PSM 1** / PSM 2

Date : 20/04/2026 **Meeting:** 1 / 2 / 3 / **4** / 5 / 6 / 7

Student
**(Meeting Minute/
Achievements/
Activities)**

: - **Case Study Identification:** Defined Alamin Clinic (Saudi Arabia) as the primary stakeholder after analyzing their specific ransomware incident

- **Problem Background:** Drafted the initial problem statement focusing on the flaws of manual server configuration and the lack of network segmentation

- **Literature Initiation:** Commenced research into the state of cloud computing in healthcare, specifically looking at studies from the early 2010s to present

Supervisor
**(Suggestion&
Comments)**

: - **Gap Analysis:** Encouraged me to focus on the "market gap" for small healthcare providers who cannot afford enterprise solutions like Epic Systems

- **Security Standards:** Advised using the HIPAA framework as a "gold-standard" benchmark for evaluating the proposed system.

- **Organizational Context:** Suggested mapping out the clinic's hierarchy to better design the future Role-Based Access Control (RBAC)

Next Meeting Plan

: **Analysis Framework:** Apply the 5W1H framework to decompose the clinic's security vulnerabilities

Complete the technology review focusing on Terraform (IaC) and DevSecOps.

Supervisor's Signature

..... **Date:**



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PSM LOG BOOK

SEMESTER: 20252026 **TYPE:** **PSM 1** / PSM 2

Date : 30/04/2026 **Meeting:** 1 / 2 / 3 / 4 / **5** / 6 / 7

Student (Meeting Minute/Achievements/Activities) : - **Chapter 1 Finalization:** Finalized the Project Aim, Objectives, and Scope, ensuring a clear distinction between in-scope (VPC, IAM, IaC) and out-of-scope (Billing, Pharmacy) modules

- **Problem:** Completed the 5W1H analysis of Alamin Clinic's vulnerabilities

- **Literature Completion:** Finalized the technology review section, linking "Shift-left" security principles to the proposed DevSecOps pipeline

- **Comparison Table:** Constructed a comparative analysis between traditional on-premise systems, OpenEMR, and the proposed cloud solution

Supervisor (Suggestion&Comments) : - **Submission Ready:** Approved the final draft of Chapters 1 and 2 for the Progress 1 evaluation

Next Meeting Plan : **Methodology Focus:** Begin drafting Chapter 3 (System Development Methodology) with a focus on the Agile phases and the Terraform implementation plan

Progress 1 Review resubmission if needed

Supervisor's Signature

Date: